

# SCP: Simulation-Based Circadian Power

May 29, 2026

**Type** Package

**Title** Simulation-Based Circadian Power Analysis with K-harmonic LRT

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**URL** <https://github.com/quythien/SCP>

**BugReports** <https://github.com/quythien/SCP/issues>

**Description** Simulation-based power analysis for circadian transcriptomics studies. Empirical parameter distributions (amplitude, noise, phase, proportion rhythmic) are estimated from a pilot dataset, and power is evaluated for rhythmicity detection and differential-rhythm tests (DR, DP, DM) across a range of study designs. Detection uses a unified cosinor F-test `detect_cosinor(K)`, with  $K = 1$  for the single-harmonic case and  $K = 2$  for the two-harmonic (ultradian) extension. A bootstrap design-grid optimiser is included for design comparison under pilot uncertainty.

**Depends** R ( $\geq 4.4.0$ )

**Imports** ggplot2,  
reshape2,  
limma,  
limorhyde,  
minpack.lm,  
nloptr,  
parallel,  
grid,  
grDevices,  
Rcpp

**LinkingTo** Rcpp,  
RcppArmadillo

**Suggests** shiny,  
shinyscreenshot,  
FMM,  
MetaCycle,  
rain,  
circacompare,  
patchwork,

cowplot,  
 gtable,  
 readxl,  
 tidyr,  
 AWFisher,  
 DODR,  
 testthat ( $\geq 3.0.0$ )

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**Encoding** UTF-8

**LazyData** false

**biocViews** Software, GeneExpression, TimeCourse, DifferentialExpression, Transcriptomics

**RoxygenNote** 7.3.2

**VignetteBuilder** knitr

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---

CircadianAnalysisOptions

*Create Analysis & Reporting Options*


---

**Description**

Create Analysis & Reporting Options

**Usage**

```
CircadianAnalysisOptions(
  alpha = 0.05,
  p.adjust.method = "BH",
  parallel.ncores = 1,
  amp.cutoff = 0,
  target_effect = 0.1,
  fdr_thresholds = c(0.01, 0.05, 0.1, 0.2),
  reference_n = 60,
  r_strata = c(0, 0.25, 0.5, 0.75, 1, 1.25, 1.5, 1.75, 2, 2.5, 3, 3.5, 4, 4.5, 5),
  strata_labels = NULL,
  phase_shifts = c(0, 0.5, 1, 2, 4, 6, 8, 10, 12),
  DCmethod = "DCP"
)
```

**Arguments**

|                 |                                                            |
|-----------------|------------------------------------------------------------|
| alpha           | Significance level for DCP pipeline                        |
| p.adjust.method | Multiple testing correction method                         |
| parallel.ncores | Number of cores for parallel DCP                           |
| amp.cutoff      | Amplitude cutoff for DCP_Rhythmicity                       |
| target_effect   | Minimum effect size to count as "interesting"              |
| fdr_thresholds  | FDR thresholds for power curves                            |
| reference_n     | Reference sample size for Panel B / phase shift plots      |
| r_strata        | Breakpoints for A/sigma stratification                     |
| strata_labels   | Labels for strata (auto-generated if NULL)                 |
| phase_shifts    | Phase shift magnitudes to sweep (for sensitivity analysis) |

**Value**

Object of class "CircadianAnalysisOptions"

---

CircadianBioOptions

*Create Biology + Differential Options*


---

## Description

Create Biology + Differential Options

## Usage

```
CircadianBioOptions(
  ngenes = 5000,
  prop_rhythmic = NULL,
  period = 24,
  lBaselineExpr = NULL,
  lBaselineExpr2 = NULL,
  lod = NULL,
  lod2 = NULL,
  amplitude = NULL,
  amplitude2 = NULL,
  sigma_rhythmic = NULL,
  paired_sigma = FALSE,
  paired_omega = FALSE,
  paired_alpha = FALSE,
  omega_rhythmic = NULL,
  alpha_rhythmic = NULL,
  omega_dist = NULL,
  alpha_dist = NULL,
  cts = NULL,
  cts2 = NULL,
  phase = "uniform",
  prop_DR = 0.15,
  prop_DP = 0.1,
  prop_DM = 0,
  phase_diff = c(-6, 6),
  amp_diff = c(0.5, 2),
  mesor_diff = c(0.5, 2),
  dp_shift_mode = c("fixed", "uniform"),
  dr_amp_scale = 1,
  dr_sigma_scale = 1,
  alpha_pilot = NULL,
  sim.seed = 12345
)
```

## Arguments

|                            |                                                                                                                                                                   |
|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>ngenes</code>        | Number of genes.                                                                                                                                                  |
| <code>prop_rhythmic</code> | Proportion of rhythmic genes. If <code>NULL</code> and a pilot dataset name is supplied to <code>lBaselineExpr</code> , the pilot's estimated proportion is used. |

|                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| period         | Circadian period in hours.                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| lBaselineExpr  | Log baseline expression. Can be a numeric scalar (constant for all genes), a numeric vector (resampled to <code>ngenes</code> ), a function <code>f(n)</code> returning <code>n</code> values, or a character string naming a built-in pilot dataset stored in <code>data/</code> . No default, must be supplied.                                                                                                                                                                                     |
| lBaselineExpr2 | As <code>lBaselineExpr</code> but for group 2 (differential analyses only). NULL uses the same distribution as group 1.                                                                                                                                                                                                                                                                                                                                                                               |
| lOD            | Log over-dispersion (noise). Same forms accepted as <code>lBaselineExpr</code> . No default, must be supplied.                                                                                                                                                                                                                                                                                                                                                                                        |
| lOD2           | As <code>lOD</code> but for group 2. NULL shares group 1 values.                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| amplitude      | Amplitude distribution for rhythmic genes. Same forms accepted as <code>lBaselineExpr</code> . No default, must be supplied.                                                                                                                                                                                                                                                                                                                                                                          |
| amplitude2     | As <code>amplitude</code> but for group 2.                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| sigma_rhythmic | Optional numeric vector of per-gene noise values for rhythmic genes (same length as <code>amplitude</code> ). When provided, amplitude and sigma are drawn jointly to preserve pilot A-sigma correlation.                                                                                                                                                                                                                                                                                             |
| paired_omega   | Logical. If TRUE and <code>omega_rhythmic</code> is supplied (same length as <code>amplitude</code> ), expand omega using the same shared gene-index used for amplitude/sigma so all three are paired per gene.                                                                                                                                                                                                                                                                                       |
| paired_alpha   | Logical. Same as <code>paired_omega</code> for <code>alpha_rhythmic</code> .                                                                                                                                                                                                                                                                                                                                                                                                                          |
| omega_rhythmic | Optional numeric vector of per-gene FMM omega values from a pilot FMM fit (same length as <code>amplitude</code> ). Used when <code>paired_omega = TRUE</code> ; ignored when <code>omega_dist</code> is also set.                                                                                                                                                                                                                                                                                    |
| alpha_rhythmic | Optional numeric vector of per-gene FMM alpha values in radians from a pilot FMM fit (same length as <code>amplitude</code> ).                                                                                                                                                                                                                                                                                                                                                                        |
| omega_dist     | Optional list specifying a distribution from which omega is drawn at simulation time, e.g. <code>list(family="beta", a=1, b=5)</code> or <code>list(family="fixed", value=0.7)</code> . Overrides <code>omega_rhythmic</code> when both are provided.                                                                                                                                                                                                                                                 |
| alpha_dist     | Optional list specifying a distribution for alpha, <code>list(family="normal", mean=0, sd_hours=2)</code> . Internally <code>sd_hours</code> is converted to radians via <code>sd_rad = sd_hours * 2*pi/24</code> . When <code>alpha_rhythmic</code> is also provided and lengths match, the perturbation is added to each gene's empirical alpha ( <code>alpha_g = alpha_rhythmic[g] + N(0, sd_rad^2)</code> ). When <code>alpha_rhythmic</code> is missing, <code>alpha_g = N(0, sd_rad^2)</code> . |
| cts            | Numeric vector of sample collection times for passive design.                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| cts2           | As <code>cts</code> for group 2.                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| phase          | Phase distribution for rhythmic genes. "uniform" (default) or the same forms as <code>lBaselineExpr</code> .                                                                                                                                                                                                                                                                                                                                                                                          |
| prop_DR        | Proportion with differential rhythmicity.                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| prop_DP        | Proportion with differential phase.                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| prop_DM        | Proportion with differential mesor (mean shift, both groups rhythmic).                                                                                                                                                                                                                                                                                                                                                                                                                                |
| phase_diff     | Range of phase shift for DP genes <code>c(min, max)</code> .                                                                                                                                                                                                                                                                                                                                                                                                                                          |

|                |                                                                                                                                                                                                                                                  |
|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| amp_diff       | Range of amplitude ratio (unused; retained for interface compatibility).                                                                                                                                                                         |
| mesor_diff     | Range of mesor shift for DM genes <code>c(min, max)</code> (additive, log-scale units).                                                                                                                                                          |
| dp_shift_mode  | "fixed" (use <code>phase_diff[2]</code> ) or "uniform" (sample uniformly within <code>phase_diff</code> range).                                                                                                                                  |
| dr_amp_scale   | Scale factor for amplitude (A) to adjust DR strength.                                                                                                                                                                                            |
| dr_sigma_scale | Scale factor for sigma to adjust DR strength.                                                                                                                                                                                                    |
| alpha_pilot    | Rhythmicity pre-screen threshold used to define the pilot's rhythmic gene set (the value at which <code>prop_rhythmic</code> and the effect-size distribution were calibrated). Stored on the object for provenance and reporting; default NULL. |
| sim.seed       | Random seed.                                                                                                                                                                                                                                     |

**Value**

Object of class "CircadianBioOptions".

---

CircadianBootstrapOptions

*Create Bootstrap Design Grid Options*

---

**Description**

Create Bootstrap Design Grid Options

**Usage**

```
CircadianBootstrapOptions(
  design_vector,
  B_values = c(4, 6, 8, 12, 24),
  m_values = NULL,
  N_values = NULL,
  nboot = 200,
  nsims_inner = 20,
  design = c("active", "passive"),
  seed = 42
)
```

**Arguments**

|               |                                                                      |
|---------------|----------------------------------------------------------------------|
| design_vector | Ordered time points to sweep over (length $\geq \max(B\_values)$ )   |
| B_values      | Number of distinct time points to test                               |
| m_values      | Replicates per time point (if NULL, derived as $\text{round}(N/B)$ ) |
| N_values      | Total per-group sample sizes to test (if NULL, derived as $B*m$ )    |
| nboot         | Number of bootstrap parameter draws (outer uncertainty loop)         |
| nsims_inner   | Simulations per bootstrap draw                                       |
| design        | "active" or "passive"                                                |
| seed          | Random seed                                                          |

**Value**

Object of class "CircadianBootstrapOptions"

---

CircadianDesignOptions  
*Create Study Design Options*

---

**Description**

Create Study Design Options

**Usage**

```
CircadianDesignOptions(
  sample_sizes = c(10, 20, 40, 60, 80, 100),
  nsims = 100,
  design = c("active", "passive"),
  cts = NULL,
  B_values = NULL,
  test_types = c("DR", "DP", "DM"),
  omega = 1,
  beta = pi
)
```

**Arguments**

`sample_sizes` Vector of sample sizes per group  
`nsims` Number of simulation replicates  
`design` "active" or "passive"  
`cts` Time-of-day distribution for passive design (numeric vector)  
`test_types` Character vector of tests to run ("DR", "DP", "DM")

**Value**

Object of class "CircadianDesignOptions"

---

CircaPower *Utility Functions for Circadian Power Analysis*

---

**Description**

Fast closed-form power calculation based on F-test. Useful for quick estimates before running full simulations. Based on CircaPower's analytical approach.

**Usage**

```
CircaPower(n = NULL, power = NULL, r = NULL, alpha = 0.05, design_factor = 0.5)
```

**Arguments**

|                            |                                               |
|----------------------------|-----------------------------------------------|
| <code>n</code>             | Sample size (NULL to solve for n)             |
| <code>power</code>         | Target power (NULL to solve for power)        |
| <code>r</code>             | Effect size ( $A/\sigma$ )                    |
| <code>alpha</code>         | Significance level                            |
| <code>design_factor</code> | Sampling design factor (0.5 for even spacing) |

**Details**

Quick analytical power for single-group rhythmicity

**Value**

List with `n`, `power`, `r`, `alpha`

**Examples**

```
# How many samples for 80% power?
CircaPower(n = NULL, power = 0.8, r = 1.5)

# What power with n=24?
CircaPower(n = 24, power = NULL, r = 1.2)
```

---

```
circaPowerApproxN80
```

*Estimate n for 80 percent power using CircaPower at median pilot r*

---

**Description**

Estimate n for 80 percent power using CircaPower at median pilot r

**Usage**

```
circaPowerApproxN80(
  bio.opts,
  alpha = 0.05,
  target_power = 0.8,
  n_search = seq(5, 500, by = 5)
)
```

**Arguments**

|                           |                                                        |
|---------------------------|--------------------------------------------------------|
| <code>bio.opts</code>     | CircadianBioOptions with amplitude and sigma_rhythmic. |
| <code>alpha</code>        | Significance level (default 0.05).                     |
| <code>target_power</code> | Target power (default 0.80).                           |
| <code>n_search</code>     | Integer vector of candidate sample sizes.              |

**Value**

Single integer – smallest n achieving target\_power, or NA if not found.



---

|                |                                                                          |
|----------------|--------------------------------------------------------------------------|
| detect_cosinor | <i>Unified cosinor rhythmicity detection (single- or multi-harmonic)</i> |
|----------------|--------------------------------------------------------------------------|

---

## Description

Canonical entry point for rhythmicity detection by K-harmonic cosinor regression.  $K = 1$  is the single-harmonic cosinor F-test, identical to [detect\\_DCP](#);  $K = 2$  adds the 12-hour second harmonic and tests the joint harmonic block. By default the unshrunk exact nested F-test is used (matching the power derivations in the paper). Returns raw p-values; the caller applies FDR control.

## Usage

```
detect_cosinor(expr, times, K = 1L, period = 24, ebayes = FALSE, mc.cores = 1L)
```

## Arguments

|                       |                                                                                                                       |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------|
| <code>expr</code>     | Gene x sample expression matrix                                                                                       |
| <code>times</code>    | Numeric time vector (length = <code>ncol(expr)</code> )                                                               |
| <code>K</code>        | Harmonic order (default 1)                                                                                            |
| <code>period</code>   | Period in hours (default 24)                                                                                          |
| <code>ebayes</code>   | For $K \geq 2$ , moderate residual variance with empirical-Bayes shrinkage (default FALSE, the unshrunk exact F-test) |
| <code>mc.cores</code> | Cores for the $K \geq 2$ fit (default 1)                                                                              |

## Value

Numeric vector of raw p-values, length `nrow(expr)`

## See Also

[detect\\_DCP](#)

---

|                                |                                                                      |
|--------------------------------|----------------------------------------------------------------------|
| <code>estCircadianParam</code> | <i>Estimate Circadian Parameters from Pilot Data (Single-Cohort)</i> |
|--------------------------------|----------------------------------------------------------------------|

---

## Description

Fits a cosinor model per gene, ranks rhythmic genes by p-value, and returns empirical parameter distributions packaged as a `CircadianBioOptions` object suitable for downstream simulation and power analysis.

**Usage**

```
estCircadianParam(
  data,
  times,
  period = 24,
  min_rhythm_pval = 0.01,
  prop_DR = 0.15,
  prop_DP = 0.1,
  prop_DM = 0,
  mesor_diff = c(0.5, 2),
  phase_diff = c(-6, 6),
  amp_diff = c(0.5, 2),
  dp_shift_mode = c("fixed", "uniform"),
  dr_amp_scale = 1,
  dr_sigma_scale = 1,
  paired_sigma = FALSE,
  sim.seed = 12345,
  verbose = TRUE
)
```

**Arguments**

|                                               |                                                                                                                                                                                                                                                                    |
|-----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>data</code>                             | Numeric matrix of pilot expression (genes x samples). A data.frame is coerced to matrix.                                                                                                                                                                           |
| <code>times</code>                            | Numeric vector of sample collection times in hours, length <code>ncol(data)</code> . The function does NOT validate units; pass hours (0 to 24 scale).                                                                                                             |
| <code>period</code>                           | Period in the same units as <code>times</code> (default 24 hours).                                                                                                                                                                                                 |
| <code>min_rhythm_pval</code>                  | Uncorrected cosinor p-value threshold for the rhythmic-candidate set (default 0.01). Used to compute <code>prop_rhythmic</code> and to select the top-K ( $K = \min(300, n_{\text{rhythmic}})$ ) genes whose amplitudes, phases, and noise drive simulation truth. |
| <code>prop_DR, prop_DP, prop_DM</code>        | Proportions of differentially-rhythmic, differentially-phased, and differentially-mesor genes (defaults 0.15, 0.10, 0.00). Capped automatically at <code>prop_rhythmic</code> .                                                                                    |
| <code>mesor_diff, phase_diff, amp_diff</code> | Numeric length-2 ranges for the differential effect-size sampling.                                                                                                                                                                                                 |
| <code>dp_shift_mode</code>                    | One of "fixed" or "uniform" (default "fixed").                                                                                                                                                                                                                     |
| <code>dr_amp_scale, dr_sigma_scale</code>     | Scaling factors for DR-class amplitude and noise distributions (default 1).                                                                                                                                                                                        |
| <code>paired_sigma</code>                     | Logical, default FALSE. If TRUE, joint (A, sigma) sampling preserves pilot correlation in downstream simulations.                                                                                                                                                  |
| <code>sim.seed</code>                         | Integer seed for downstream simulation (default 12345).                                                                                                                                                                                                            |
| <code>verbose</code>                          | Logical, default TRUE. Print summary at end.                                                                                                                                                                                                                       |

**Value**

A `CircadianBioOptions` S3 object usable directly with `runSimsSingleCohort()` or `runDifferentialPower()`.

**See Also**

[estCircadianParamFMM](#) for FMM-aware pre-screen, [estCircadianParamTwoGroup](#) for two-group setup, [prepCircadianData](#) for normalisation upstream.

**Examples**

```
## Not run:
set.seed(1)
n_genes <- 200; n_samples <- 30
times <- rep(seq(0, 22, by = 2), length.out = n_samples)
expr <- matrix(rnorm(n_genes * n_samples), nrow = n_genes)
for (g in 1:20) expr[g, ] <- expr[g, ] + 2 * cos(2*pi*times/24)
bio <- estCircadianParam(expr, times, period = 24, verbose = FALSE)

## End(Not run)
```

---

```
estCircadianParam2H
```

*Estimate CircadianBioOptions Under the Two-Harmonic Cosinor Model*

---

**Description**

Two-harmonic analog of [estCircadianParam](#). Fits the regression

$$y = \mu + a_1 \cos(\omega_0 t) + b_1 \sin(\omega_0 t) + a_2 \cos(2\omega_0 t) + b_2 \sin(2\omega_0 t) + \epsilon$$

to each gene by OLS, tests the joint rhythmicity null  $H_0 : a_1 = b_1 = a_2 = b_2 = 0$  on 4 d.f., and returns a `CircadianBioOptions` object whose 5-way joint empirical distribution  $F_{(\mu, A_1, \phi_1, A_2, \phi_2, \sigma)}$  is preserved by gene-paired index draws downstream.

The returned object carries the standard 1-harmonic fields (`amplitude`, `phase`, `sigma_rhythmic`) holding the 1st-harmonic  $A_{g,1}$ ,  $\phi_{g,1}$ , and  $\sigma_g$  respectively, and three extra fields attached to the list after construction:

- `amplitude2`: 2nd-harmonic amplitudes  $A_{g,2}$
- `phase2`: 2nd-harmonic acrophases  $\phi_{g,2}$  in hours  $[0, period/2)$
- `paired_2h`: TRUE flag (downstream simulators read this to switch on the 2-harmonic generative model).

These extras live alongside `$diagnostics`; the constructor is not modified.

**Usage**

```
estCircadianParam2H(
  data,
  times,
  period = 24,
  min_rhythm_pval = 0.01,
  top_k = 500,
  prop_DR = 0.15,
  prop_DP = 0.1,
```

```

prop_DM = 0,
mesor_diff = c(0.5, 2),
phase_diff = c(-6, 6),
amp_diff = c(0.5, 2),
dp_shift_mode = c("fixed", "uniform"),
dr_amp_scale = 1,
dr_sigma_scale = 1,
paired_sigma = FALSE,
sim.seed = 12345,
verbose = TRUE
)

```

### Arguments

|                                               |                                                                                                                                                                      |
|-----------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>data</code>                             | Numeric matrix of pilot expression (genes x samples).                                                                                                                |
| <code>times</code>                            | Numeric vector of sample collection times in hours, length <code>ncol(data)</code> .                                                                                 |
| <code>period</code>                           | Period in the same units as <code>times</code> (default 24 hours).                                                                                                   |
| <code>min_rhythm_pval</code>                  | P-value threshold for the joint 4-d.f. F-test (default 0.01). Genes with $p < \text{this}$ define the candidate set.                                                 |
| <code>top_k</code>                            | Cap on number of genes contributing to the joint $(A_1, \phi_1, A_2, \phi_2, \sigma)$ distribution (default 300), ranked by F-test p-value.                          |
| <code>prop_DR, prop_DP, prop_DM</code>        | Differential proportions (defaults 0.15, 0.10, 0.00). Forwarded unchanged to <code>CircadianBioOptions</code> ; note the K=2 extension itself is single-cohort only. |
| <code>mesor_diff, phase_diff, amp_diff</code> | Differential effect-size ranges.                                                                                                                                     |
| <code>dp_shift_mode</code>                    | "fixed" or "uniform" (default "fixed").                                                                                                                              |
| <code>dr_amp_scale, dr_sigma_scale</code>     | DR scale factors (default 1).                                                                                                                                        |
| <code>paired_sigma</code>                     | Logical (default FALSE). Forwarded to <code>CircadianBioOptions</code> .                                                                                             |
| <code>sim.seed</code>                         | Integer seed (default 12345).                                                                                                                                        |
| <code>verbose</code>                          | Logical (default TRUE). Print diagnostic summary.                                                                                                                    |

### Value

A `CircadianBioOptions` object with extra fields `$amplitude2`, `$phase2`, `$paired_2h`, and a `$diagnostics` list summarising the 2H fit (median  $A_2/A_1$  ratio, fraction of genes with significant 2nd-harmonic block, etc.).

### Methodological notes

1. Pilot summary is a 5-way joint empirical distribution. When the simulator draws a rhythmic gene, the same index  $j$  is used across  $(A_1, \phi_1, A_2, \phi_2, \sigma)$ , preserving the cross-parameter dependence observed in the pilot.
2. No 2nd-harmonic significance threshold is applied: ALL top-K genes passing the joint 4-d.f. F-test contribute, even those whose  $A_2$  is essentially noise. The empirical  $F_{A_2}$  will have a noise spike near zero, which is fine.
3. K=2 extension is single-cohort only – the differential simulator (`simCircadianDiff`) is not extended.

**See Also**

[estCircadianParam](#) for the 1-harmonic case, [simCircadianSingleCohort2H](#) for the matching simulator.

---

```
estCircadianParamTwoGroup
```

*Estimate a two-group circadian pilot for differential power*

---

**Description**

Fits cosinor models to two groups of pilot data and assembles the two-group pilot summary used by the differential power simulator, encoding differential rhythmicity (DR), differential phase (DP), and differential MESOR (DM) structure.

**Usage**

```
estCircadianParamTwoGroup(
  data_1,
  data_2,
  times_1,
  times_2,
  period = 24,
  min_rhythm_pval = 0.01,
  phase_shift_threshold = 2,
  prop_DM = NULL,
  mesor_diff = NULL,
  dp_shift_mode = c("uniform", "fixed"),
  paired_sigma = FALSE,
  sim.seed = 12345,
  verbose = TRUE
)
```

**Arguments**

|                                    |                                                                    |
|------------------------------------|--------------------------------------------------------------------|
| <code>data_1, data_2</code>        | Gene-by-sample expression matrices for the two groups.             |
| <code>times_1, times_2</code>      | Numeric collection times for each group.                           |
| <code>period</code>                | Period in hours (default 24).                                      |
| <code>min_rhythm_pval</code>       | Marginal p-value screen defining the rhythmic pilot set.           |
| <code>phase_shift_threshold</code> | Minimum phase difference (hours) for the DP class.                 |
| <code>prop_DM</code>               | Optional target proportion of differential-MESOR genes.            |
| <code>mesor_diff</code>            | Optional MESOR difference for the DM component.                    |
| <code>dp_shift_mode</code>         | Differential-phase shift model, "uniform" or "fixed".              |
| <code>paired_sigma</code>          | Preserve the paired (amplitude, noise) relationship across groups. |
| <code>sim.seed</code>              | Random seed.                                                       |
| <code>verbose</code>               | Print progress.                                                    |

**Value**

A two-group CircadianBioOptions pilot summary.

---

|             |                                 |
|-------------|---------------------------------|
| launchShiny | <i>Launch the SCP Shiny app</i> |
|-------------|---------------------------------|

---

**Description**

Opens the bundled simulation-based power-analysis app in the user's default browser. The app lets the user pick a bundled pilot, set target power and FDR, and read off the recommended sample size for single-cohort rhythmicity detection.

**Usage**

```
launchShiny(...)
```

**Arguments**

... Additional arguments passed to shiny::runApp().

**Value**

Invisibly returns NULL. Called for its side effect of launching the Shiny app.

**Examples**

```
## Not run:
  launchShiny()

## End(Not run)
```

---

|                     |                                                                   |
|---------------------|-------------------------------------------------------------------|
| makeAdaptiveRStrata | <i>Compute adaptive r-strata breakpoints with fixed bin width</i> |
|---------------------|-------------------------------------------------------------------|

---

**Description**

Generates breakpoints from 0 to ceiling(r\_max / bin\_width) \* bin\_width, stepping by bin\_width, then appends Inf. Every bin width is identical, but the number of bins adapts to the empirical r range of the pilot data.

**Usage**

```
makeAdaptiveRStrata(bio.opts, bin_width = 0.25, r_min_pct = 0)
```

**Arguments**

|           |                                                                                    |
|-----------|------------------------------------------------------------------------------------|
| bio.opts  | CircadianBioOptions with amplitude and sigma_rhythmic.                             |
| bin_width | Width of each r bin (default 0.25).                                                |
| r_min_pct | Lower percentile of pilot r used as range floor (default 0 = always start from 0). |

**Value**

Numeric vector of breakpoints starting at 0 and ending at Inf.

---

|        |                                                                          |
|--------|--------------------------------------------------------------------------|
| npower | <i>Find the sample size needed to achieve a target genome-wide power</i> |
|--------|--------------------------------------------------------------------------|

---

**Description**

Works on output from either `runSimsSingleCohort()` or `runSimsDiff()`. Recomputes marginal power at the requested FDR from raw simulation output, then returns the smallest grid N that meets the target (`n_grid`) and, when `interpolate = TRUE` (the default), a linearly interpolated estimate between the two bracketing grid points (`n`).

**Usage**

```
npower(
  res,
  target_power = 0.8,
  fdr = 0.05,
  endpoint = NULL,
  p.adjust.method = "BH",
  interpolate = TRUE
)
```

**Arguments**

|                              |                                                                                                                                                                                                                                                                |
|------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>res</code>             | Output from <code>runSimsSingleCohort()</code> or <code>runSimsDiff()</code> .                                                                                                                                                                                 |
| <code>target_power</code>    | Numeric in (0, 1). Target marginal power (default 0.80).                                                                                                                                                                                                       |
| <code>fdr</code>             | FDR threshold (default 0.05).                                                                                                                                                                                                                                  |
| <code>endpoint</code>        | For <code>runSimsDiff()</code> output: one of "DR", "DP", "DM". Ignored for single-cohort output.                                                                                                                                                              |
| <code>p.adjust.method</code> | Multiple-testing correction (default "BH"). Only applied for single-cohort output (differential uses pre-computed FDR arrays).                                                                                                                                 |
| <code>interpolate</code>     | Logical. If <code>TRUE</code> (default), linearly interpolates between the two adjacent grid points to return a more precise <code>n</code> estimate. Set to <code>FALSE</code> to revert to the grid-snapped <code>n_grid</code> value (backward-compatible). |

**Value**

An object of class "npower" (a list) with:

|                                          |                                                                                                                                                                                       |
|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>n</code>                           | Recommended sample size: <code>n_interp</code> when <code>interpolate = TRUE</code> , otherwise <code>n_grid</code> . NA if target power is never reached.                            |
| <code>n_grid</code>                      | Smallest sample size in the simulation grid achieving target power, or NA.                                                                                                            |
| <code>n_interp</code>                    | Linearly interpolated sample size between the two bracketing grid points (a fractional N rounded up). NA when <code>interpolate = FALSE</code> or when target power is never reached. |
| <code>power</code>                       | Named numeric vector: mean power at each grid sample size.                                                                                                                            |
| <code>target_power, fdr, endpoint</code> | Echo of inputs.                                                                                                                                                                       |

---

```
plotBvsMPower =====
plot_bvsm.R,  B   vs   m   Trade-off   Plot   (Figure 3)
=====
```

---

## Description

Produces a faceted power-vs-N figure for each dataset in the result, with one line per B value. SD bars (+/-1 SD across simulation replicates) are drawn at each N. B-invariance appears as overlapping lines.

## Usage

```
plotBvsMPower (
  x,
  nsims = 30L,
  alpha2 = 0,
  fdr_power = 0.8,
  dataset_labels = NULL,
  output_file = NULL,
  width = 14,
  height = 5
)
```

## Arguments

|                |                                                                                                                                          |
|----------------|------------------------------------------------------------------------------------------------------------------------------------------|
| x              | An SCPSingleResult from runSingleCohortGrid(), or a tidy data.frame with columns N, B, power, power_se, dataset (and optionally alpha2). |
| nsims          | Number of simulation replicates used (to convert SE -> SD). Default 30.                                                                  |
| alpha2         | Which alpha2 value to plot (default 0 = pure cosinor).                                                                                   |
| fdr_power      | Reference power level for dashed line (default 0.80).                                                                                    |
| dataset_labels | Named character vector mapping dataset codes to display labels. If NULL, uses dataset column values directly.                            |
| output_file    | Path to save PDF. NULL = current graphics device.                                                                                        |
| width          | PDF width in inches (default 14).                                                                                                        |
| height         | PDF height in inches (default 5).                                                                                                        |

## Details

Plots DCP power vs N for multiple B values, demonstrating B-invariance. Input: SCPSingleResult from runSingleCohortGrid() or a tidy data.frame with columns N, B, alpha2, power, power\_se. Plot B vs m Trade-off Power Grid

## Value

Invisibly returns the ggplot object.



---

|               |                                          |
|---------------|------------------------------------------|
| plotDiffPower | <i>Plot differential circadian power</i> |
|---------------|------------------------------------------|

---

## Description

Default layout is marginal power vs sample size only (one column per endpoint, one row per comparison). The effect-size-stratified panels (power and true-discoveries by  $r$ ) are opt-in via `stratified = TRUE`, which restores the original 3-column-per-row layout.

## Usage

```
plotDiffPower(
  res_list,
  comp_labels = names(res_list),
  endpoints = c("DR", "DP", "DM"),
  fdr_thresholds = c(0.01, 0.05, 0.1, 0.2),
  panel_fdr = 0.05,
  vline_power = 0.8,
  vline_fdr = 0.2,
  display_sizes = NULL,
  stratified = FALSE,
  r_break_width = 0.25,
  r_max = 5,
  r_display_max = NULL,
  main_title = "Differential Circadian Power Analysis",
  out_pdf = NULL,
  width = NULL,
  height = NULL
)
```

## Arguments

|                             |                                                                                                                                                                                                                                                                        |
|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>res_list</code>       | Named list of length 2, each element output of <code>runSimsDiff()</code> .                                                                                                                                                                                            |
| <code>comp_labels</code>    | Character(2), display labels for the two comparisons.                                                                                                                                                                                                                  |
| <code>endpoints</code>      | Which endpoints to include (default <code>c("DR", "DP", "DM")</code> ).                                                                                                                                                                                                |
| <code>fdr_thresholds</code> | FDR levels for panel A (default <code>c(0.01, 0.05, 0.10, 0.20)</code> ).                                                                                                                                                                                              |
| <code>display_sizes</code>  | Sample sizes to show in all panels (NULL = all).                                                                                                                                                                                                                       |
| <code>stratified</code>     | Logical (default FALSE). If FALSE, draws only the marginal power-vs-sample-size panel per endpoint (1 x <code>n_endpoints</code> layout per comparison). If TRUE, also draws power-by- $r$ and true-discoveries-by- $r$ strata panels in the original 3-column layout. |
| <code>r_break_width</code>  | Width of $r$ -strata bins (default 0.25). Ignored if <code>stratified = FALSE</code> .                                                                                                                                                                                 |
| <code>out_pdf</code>        | Path for PDF. If NULL, plots to current device.                                                                                                                                                                                                                        |
| <code>width</code>          | PDF width in inches. Defaults adapt to stratified.                                                                                                                                                                                                                     |
| <code>height</code>         | PDF height in inches. Defaults adapt to stratified.                                                                                                                                                                                                                    |

**Value**

Invisibly, list of panel data per comparison/endpoint.

---

```
plotSingleCohortPower
```

*Plot single-cohort circadian power (3 panels)*

---

**Description**

Plot single-cohort circadian power (3 panels)

**Usage**

```
plotSingleCohortPower(
  res,
  out_pdf = NULL,
  title = "",
  panel_a_res = NULL,
  panels = c("A", "B", "C"),
  fdr_thresholds = c(0.01, 0.05, 0.1, 0.2),
  panel_fdr = 0.05,
  vline_power = 0.8,
  vline_fdr = 0.2,
  fdr = NULL,
  p.adjust.method = "BH",
  reference_n = NULL,
  display_sizes = NULL,
  r_max = 5,
  cex_main = 1.5,
  cex_lab = 1.6,
  cex_axis = 1.45,
  vertical = FALSE,
  panel_c_legend = FALSE,
  width = 15,
  height = 5.5
)
```

**Arguments**

|                              |                                                                                                                                                                                                                                                                                             |
|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>res</code>             | Output from <code>runSimsSingleCohort()</code> .                                                                                                                                                                                                                                            |
| <code>out_pdf</code>         | Path for PDF output. If <code>NULL</code> , plots to current device.                                                                                                                                                                                                                        |
| <code>title</code>           | Overall figure title (default empty).                                                                                                                                                                                                                                                       |
| <code>panel_a_res</code>     | Optional alternate result object used for the Panel A marginal-power computation only. If <code>NULL</code> (default), Panel A is computed from <code>res</code> . Useful when the marginal summary should reflect a different empirical assumption than the effect-size-stratified panels. |
| <code>fdr_thresholds</code>  | FDR levels for panel A (default <code>c(0.01,0.05,0.10,0.20)</code> ).                                                                                                                                                                                                                      |
| <code>p.adjust.method</code> | Correction method (default "BH").                                                                                                                                                                                                                                                           |

|             |                                                              |
|-------------|--------------------------------------------------------------|
| reference_n | Reference sample size for annotations (default NULL = none). |
| width       | PDF width in inches (default 15).                            |
| height      | PDF height in inches (default 5.5).                          |

**Value**

Invisibly returns list of data used per panel.

---

|                 |                                                      |
|-----------------|------------------------------------------------------|
| recommendDesign | <i>B vs m design study – analytical + simulation</i> |
|-----------------|------------------------------------------------------|

---

**Description**

Full B vs m study orchestrator. Three sequential steps:

**Usage**

```
recommendDesign(
  bio.opts,
  design.opts,
  analysis.opts,
  methods = c("DCP", "JTK", "RAIN", "MH"),
  target_power = 0.8,
  mode = c("single", "differential"),
  run_simulation = TRUE,
  prior_result = NULL,
  alpha2 = 0,
  alpha3 = 0,
  mc.cores = 1L,
  plot = TRUE,
  output_file = NULL,
  verbose = TRUE
)
```

**Arguments**

|                |                                                                                                                                    |
|----------------|------------------------------------------------------------------------------------------------------------------------------------|
| bio.opts       | CircadianBioOptions from estCircadianParam() or estCircadianParamTwoGroup()                                                        |
| design.opts    | CircadianDesignOptions with sample_sizes and B_values                                                                              |
| analysis.opts  | CircadianAnalysisOptions                                                                                                           |
| methods        | Detection methods to compare. Single-cohort: c("DCP","JTK","RAIN","MH"). Differential: c("DCP","CircaCompare","LimoRhyde","DODR"). |
| target_power   | Target power for recommendation (default 0.80)                                                                                     |
| mode           | "single" or "differential" (default "single")                                                                                      |
| run_simulation | TRUE = run full simulation after analytical step. FALSE = analytical (CircaPower) only.                                            |
| prior_result   | A previous runSingleCohortGrid()/runDifferentialPower() result to reuse instead of re-running simulation.                          |

|             |                                           |
|-------------|-------------------------------------------|
| alpha2      | 2nd-harmonic deviation (scalar or vector) |
| alpha3      | 3rd-harmonic deviation (scalar or vector) |
| mc.cores    | Parallel cores for simulation step        |
| plot        | Auto-plot final recommendation            |
| output_file | PDF path (NULL = screen)                  |
| verbose     | Print progress                            |

### Details

1. Print method guidance table (printMethodGuidance) 2. Analytical: CircaPower closed-form power at each N (DCP; B-invariant). For JTK/RAIN/MH no closed form exists – simulation is required. 3. Simulation: run runSingleCohortPower() or runDifferentialPower(), or absorb a prior\_result from a previous call to skip re-running.

### Value

SCPRecommendResult with: \$guidance – data.frame: method guidance table \$analytical\_df – data.frame[method, N, power\_analytical] (DCP only) \$simulation – runSingleCohortGrid() or runDifferentialPower() result (NULL if not run) \$recommendation – data.frame[method, optimal\_B, n\_target, note]

---

```
runBootstrapDesignGrid
```

*Run bootstrap design grid simulation*

---

### Description

Estimates uncertainty in power across (N, B) design configurations by bootstrapping parameter sets from pilot data.

### Usage

```
runBootstrapDesignGrid(
  pilot_data,
  pilot_times,
  boot.opts,
  analysis.opts,
  bio_diff.opts,
  pilot_data_2 = NULL,
  pilot_times_2 = NULL,
  mode = NULL,
  methods = "DCP",
  test_types = c("DR", "DP", "DM"),
  alpha2 = 0,
  alpha3 = 0,
  min_rhythm_pval = 0.01,
  verbose = TRUE,
  mc.cores = 1L
)
```

**Arguments**

|                              |                                                                                                                                                                 |
|------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>pilot_data</code>      | Genes x samples matrix (pilot experiment).                                                                                                                      |
| <code>pilot_times</code>     | Numeric vector of sample time points, length = <code>ncol(pilot_data)</code> .                                                                                  |
| <code>boot.opts</code>       | <code>CircadianBootstrapOptions</code> from <code>CircadianBootstrapOptions()</code> .<br>Must set <code>B_values</code> to a single value for passive designs. |
| <code>analysis.opts</code>   | <code>CircadianAnalysisOptions</code> .                                                                                                                         |
| <code>bio_diff.opts</code>   | <code>CircadianBioOptions</code> from <code>estCircadianParam()</code> . Required even in mode = "single", differential proportion fields are read from here.   |
| <code>pilot_data_2</code>    | Optional group-2 pilot matrix for mode = "differential".                                                                                                        |
| <code>pilot_times_2</code>   | Optional group-2 pilot times.                                                                                                                                   |
| <code>mode</code>            | "single" or "differential".                                                                                                                                     |
| <code>methods</code>         | Detection method(s): "DCP" (default), "JTK", "RAIN".                                                                                                            |
| <code>test_types</code>      | Endpoints for differential mode: any of "DR", "DP", "DM".                                                                                                       |
| <code>alpha2</code>          | Second-harmonic coefficient (default 0).                                                                                                                        |
| <code>alpha3</code>          | Third-harmonic coefficient (default 0).                                                                                                                         |
| <code>min_rhythm_pval</code> | P-value threshold for pilot rhythmicity (default 0.01).                                                                                                         |
| <code>verbose</code>         | Print progress (default TRUE).                                                                                                                                  |
| <code>mc.cores</code>        | Parallel cores for inner simulation loop (default 1).                                                                                                           |

**Value**

Named list with 15 elements:

|                              |                                                                                                |
|------------------------------|------------------------------------------------------------------------------------------------|
| <code>N_values</code>        | Candidate sample sizes tested.                                                                 |
| <code>B_values</code>        | Candidate time-point counts tested.                                                            |
| <code>m_matrix</code>        | Matrix of total samples (N x B).                                                               |
| <code>nboot</code>           | Number of bootstrap replicates.                                                                |
| <code>design</code>          | Design type ("active" or "passive").                                                           |
| <code>test_types</code>      | Endpoints evaluated.                                                                           |
| <code>fdr_threshold</code>   | FDR threshold used.                                                                            |
| <code>boot_power</code>      | 4-D array [ <code>nboot</code> x <code>n_N</code> x <code>n_B</code> x <code>n_tests</code> ]. |
| <code>power_mean</code>      | Mean power array [ <code>n_N</code> x <code>n_B</code> x <code>n_tests</code> ].               |
| <code>power_se</code>        | Bootstrap SE array, same shape as <code>power_mean</code> .                                    |
| <code>power_ci_lo</code>     | 2.5th percentile CI array.                                                                     |
| <code>power_ci_hi</code>     | 97.5th percentile CI array.                                                                    |
| <code>optimal_B</code>       | Integer vector (length <code>n_N</code> ): B with highest mean power at each N.                |
| <code>optimal_B_ci_lo</code> | Bootstrap 2.5th percentile of optimal B.                                                       |
| <code>optimal_B_ci_hi</code> | Bootstrap 97.5th percentile of optimal B.                                                      |

---

```
runDifferentialPower
```

*Run two-group differential circadian power analysis*

---

## Description

Sweeps  $N \times \alpha_2 \times \alpha_3 \times \text{method} \times \text{test\_type}$  and returns a tidy power table. Automatically calls `plot()` and `npower()` unless `plot=FALSE`.

## Usage

```
runDifferentialPower(
  bio.opts,
  design.opts,
  analysis.opts,
  methods = "DCP",
  test_types = c("DR", "DP", "DM"),
  alpha2 = 0,
  alpha3 = 0,
  mc.cores = 1L,
  plot = TRUE,
  output_file = NULL,
  verbose = TRUE
)
```

## Arguments

|                            |                                                                                               |
|----------------------------|-----------------------------------------------------------------------------------------------|
| <code>bio.opts</code>      | CircadianBioOptions from <code>estCircadianParamTwoGroup()</code>                             |
| <code>design.opts</code>   | CircadianDesignOptions ( <code>sample_sizes</code> , <code>design</code> , <code>cts</code> ) |
| <code>analysis.opts</code> | CircadianAnalysisOptions                                                                      |
| <code>methods</code>       | Any of "DCP", "CircaCompare", "LimoRhyde", "DODR"                                             |
| <code>test_types</code>    | Any of "DR", "DP", "DM" (silently NA if method lacks support)                                 |
| <code>alpha2</code>        | Scalar or vector swept for both groups                                                        |
| <code>alpha3</code>        | Scalar or vector swept for both groups                                                        |
| <code>mc.cores</code>      | Parallel cores                                                                                |
| <code>plot</code>          | Auto-plot on completion                                                                       |
| <code>output_file</code>   | PDF path (NULL = screen)                                                                      |
| <code>verbose</code>       | Print progress                                                                                |

## Value

List (class `SCPDiffResult`) from `runSimsDiff()`: `pval_DR`, `fdr_DR`, `pval_DP`, `fdr_DP`, `pval_DM`, `fdr_DM`—3-D arrays [`ngenes`  $\times$  `n_sizes`  $\times$  `nsims`]; plus `diff_type`, `effectsize`, `sample_sizes`, `nsims`. Pass to `plotDiffPower()` or `npower(..., endpoint="DR")`.

---

runSimsSingleCohort

*Simulation-Based Power for Single-Cohort Rhythmicity Detection*


---

## Description

Extends the closed-form CircaPower approach by running full simulations: for each sample size and simulation replicate, generates data from the empirical pilot parameter distributions, applies the cosinor F-test per gene, applies BH correction, and aggregates empirical power and FDR.

## Usage

```
runSimsSingleCohort (
  bio.opts,
  design.opts,
  analysis.opts,
  method = "cosinor",
  K = 1L,
  harmonics = c(0, 0),
  verbose = TRUE,
  mc.cores = 1L,
  progress = NULL
)
```

## Arguments

|               |                                                                                                                                                                                                           |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| bio.opts      | CircadianBioOptions – pilot parameter distributions. Build with <code>estCircadianParam()</code> .                                                                                                        |
| design.opts   | CircadianDesignOptions – sample sizes, nsims, design.                                                                                                                                                     |
| analysis.opts | CircadianAnalysisOptions – alpha, p.adjust.method.                                                                                                                                                        |
| method        | Detection method: "DCP" (1-harmonic cosinor F-test), "JTK", "RAIN", "MH" (multi-harmonic), or "FMM" (K-harmonic F-test motivated by the FMM Fourier expansion; default detector for non-cosinor signals). |
| K             | Integer. Number of harmonics for method = "FMM" (default 2). K=1 is equivalent to DCP; K=2 captures the 1st and 2nd Fourier harmonics; K=3+ for very sharp peaks.                                         |
| harmonics     | Numeric length-2: <code>c(alpha2, alpha3)</code> harmonic coefficients (default <code>c(0, 0)</code> ).                                                                                                   |
| verbose       | Print progress (default TRUE).                                                                                                                                                                            |
| mc.cores      | Parallel cores (default 1).                                                                                                                                                                               |

## Details

Simulation-based alternative to the closed-form CircaPower formula: works for any design (active or passive) and any pilot parameter distribution.

**Value**

List with:

```

marginal_power      Matrix [sample_sizes x nsims]
marginal_FDR        Matrix [sample_sizes x nsims]
marginal_TD         Matrix [sample_sizes x nsims]
marginal_FD         Matrix [sample_sizes x nsims]
marginal_alpha      Matrix [sample_sizes x nsims] – empirical type-I error
pvalues             Array [sample_sizes x ngenes x nsims] – raw p-values
r_values_list       List[[sample_size]][[sim]] – per-gene A/sigma
strat_power         Array [sample_sizes x n_strata x nsims]
strat_TD           Array [sample_sizes x n_strata x nsims]
strat_FD           Array [sample_sizes x n_strata x nsims]
strat_n_targets     Array [sample_sizes x n_strata x nsims]
strat_n_nulls       Array [sample_sizes x n_strata x nsims]
strat_FDR           Array [sample_sizes x n_strata x nsims]
strat_alpha         Array [sample_sizes x n_strata x nsims]
r_strata            Stratum breakpoints (from analysis.opts)
strata_labels       Stratum labels
n0_circapower       CircaPower n80 estimate from pilot median r
sample_sizes        Vector of sample sizes
nsims               Number of simulation replicates

```

**Examples**

```

## Not run:
bio <- estCircadianParam(expr, times)
dopt <- CircadianDesignOptions(sample_sizes = c(20, 40, 80), nsims = 50)
aopt <- CircadianAnalysisOptions(alpha = 0.05)
res <- runSimsSingleCohort(bio, dopt, aopt)
rowMeans(res$marginal_power) # mean power at each N

## End(Not run)

```



---

|                |                                         |
|----------------|-----------------------------------------|
| scp_load_pilot | <i>Load a Bundled SCP Pilot Dataset</i> |
|----------------|-----------------------------------------|

---

## Description

Loads one standardized pilot object from the bundled pilot database. Each pilot is a pre-fit `CircadianBioOptions` S3 object (see [estimate\\_circadian\\_params](#)) and can be passed directly to `runSimsSingleCohort()` or `runDifferentialPower()`.

## Usage

```
scp_load_pilot(
  species,
  dataset,
  tissue,
  condition = NULL,
  alpha_pilot = 0.01,
  adjust = c("none", "BH"),
  K = 1,
  paired_sigma = TRUE
)
```

## Arguments

|              |                                                                                                                                                                                                                                                                                                                                                                                                                  |
|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| species      | Character, length 1. One of "human", "baboon", "other_primate", "mouse", "rat".                                                                                                                                                                                                                                                                                                                                  |
| dataset      | Character, length 1. Manifest dataset field (typically GEO accession or short label, e.g. "GTEx", "GSE160521").                                                                                                                                                                                                                                                                                                  |
| tissue       | Character, length 1. Tissue or region label as recorded in the manifest (e.g. "Liver", "NAc").                                                                                                                                                                                                                                                                                                                   |
| condition    | Character, length 1, optional. Condition arm (e.g. "Control", "All"). If NULL and the (species, dataset, tissue) triple identifies more than one row, the function errors and prints the available conditions.                                                                                                                                                                                                   |
| alpha_pilot  | Numeric in $(0, \text{cap}]$ . Rhythmicity threshold used to re-derive <code>prop_rhythmic</code> and the amplitude/phase/sigma distributions from the pilot's stored per-gene rhythm table. Default 0.01 matches the build-time threshold. Only takes effect for pilots rebuilt with a <code>rhythm_fit</code> table; older pilots ignore it (with a warning) and return their frozen build-time distributions. |
| adjust       | Multiple-testing adjustment applied to the per-gene p-values before thresholding at <code>alpha_pilot</code> : "none" (raw cosinor p, default) or "BH" (Benjamini-Hochberg q-value).                                                                                                                                                                                                                             |
| K            | Harmonic order of the requested pilot: 1 (single-harmonic cosinor, default) or 2 (two-harmonic). For $K = 2$ the two-harmonic variant of the pilot ( <code>&lt;file&gt;_2H.rds</code> ) is loaded; an error is raised if no such variant has been built for the pilot.                                                                                                                                           |
| paired_sigma | Logical (default TRUE). Keep the per-gene (A, sigma) pairing so the simulator draws amplitude and noise from the same pilot gene, preserving the realistic narrow $\tilde{r} = A/\sigma$ distribution (the marginal-power / Fig 1-2 Panel A convention). FALSE decouples sigma from amplitude, giving the wide $\tilde{r}$ used by the effect-size-stratified panels (Fig 1-2 Panels B/C).                       |

**Value**

A `CircadianBioOptions` object with empirical distributions of mesor, log-sigma, amplitude, phase, and proportion rhythmic, ready for downstream simulation.

**Examples**

```
## Not run:
bio <- scp_load_pilot("human", "GTEEx", "Liver")
bio <- scp_load_pilot("human", "GTEEx", "Liver", alpha_pilot = 0.05)
bio <- scp_load_pilot("human", "GTEEx", "Liver", alpha_pilot = 0.1, adjust = "BH")
bio <- scp_load_pilot("human", "GSE160521", "NAc", "Control")

## End(Not run)
```

---

 scp\_pilots

---

*List Bundled SCP Pilot Datasets*


---

**Description**

Reads the bundled pilot manifest (`inst/extdata/pilots/manifest.csv`) and returns it as a `data.frame`. Optionally filters by species and ingestion status so a user can quickly enumerate the pilots they actually have available.

**Usage**

```
scp_pilots(species = NULL, status = "ingested")
```

**Arguments**

|         |                                                                                                                                                                                                                             |
|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| species | Optional character. One or more of "human", "baboon", "other_primate", "mouse", "rat". NULL (default) returns all species.                                                                                                  |
| status  | Character. Filter on the status column. Defaults to "ingested" (datasets actually present on disk). Use NULL or "all" to return every row, including "pending", "verify_failed", "controlled", and "ingest_failed" entries. |

**Value**

A `data.frame` with one row per (dataset, tissue, condition) pilot. Columns: species, dataset, tissue, condition, n, ngenes, design, assay, sample\_type, accession, citation, license, file, status.

**Examples**

```
## Not run:
scp_pilots()
scp_pilots(species = "human")
scp_pilots(status = "all")

## End(Not run)
```

---

 scp\_pilot\_search      *Search the SCP Pilot Database*


---

### Description

Convenience filter over the pilot manifest. Combines free-text substring matching against dataset, tissue, citation, and accession fields with optional species and tissue-regex filters and an in-vivo vs cell-line restriction. Useful when a user wants to pick a pilot whose design (tissue, sample size, signal strength) matches their planned study.

### Usage

```
scp_pilot_search(
  query = NULL,
  species = NULL,
  tissue_pattern = NULL,
  sample_type = "in_vivo"
)
```

### Arguments

|                |                                                                                                                                |
|----------------|--------------------------------------------------------------------------------------------------------------------------------|
| query          | Character, optional. Case-insensitive substring matched against dataset, tissue, accession, citation. NULL matches everything. |
| species        | Character, optional. Restrict to these species.                                                                                |
| tissue_pattern | Character, optional. Regular expression matched against tissue (case-insensitive).                                             |
| sample_type    | Character. Default "in_vivo". Use NULL to include cell-line pilots as well.                                                    |

### Value

A data.frame: rows of the manifest passing every active filter.

### Examples

```
## Not run:
  scp_pilot_search("liver")
  scp_pilot_search(species = "mouse", tissue_pattern = "brain|cortex")
  scp_pilot_search(sample_type = NULL)    # include cell lines

## End(Not run)
```

---

 scp\_pilot\_tod      *Get the time-of-day (TOD) vector for a bundled SCP pilot*


---

### Description

Returns the per-sample sampling times for a pilot, in hours mod 24. Prefers the raw p\$times field when the pilot rds preserved it; otherwise parses the manifest tod\_phases column and returns the unique sampling phases.

**Usage**

```
scp_pilot_tod(species, dataset, tissue, condition = NULL)
```

**Arguments**

```
species, dataset, tissue, condition  
manifest tuple identifying the pilot.
```

**Value**

Numeric vector of times in hours (mod 24); NULL if no TOD information is available for this pilot.

**Examples**

```
## Not run:  
tod <- scp_pilot_tod("baboon", "GSE98965", "SCN", "All")  
hist(tod, breaks = seq(0, 24, by = 1))  
  
## End(Not run)
```

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